

Introduction to Analysis Homework 2

February, 27, 2025

1. (10 pt) Prove or disprove the following statements.

(a) If f is Riemann integrable on $[a, b]$, then $F(x) = \int_a^x f(t)dt$ is differentiable on $[a, b]$.

(b) If F is differentiable on $[a, b]$, then $\int_a^x F'(t)dt = F(x) - F(a)$.

2. (10 pt) Suppose that f, g are Riemann integrable on $[a, b]$ with $g(x) \geq 0$ for all $x \in [a, b]$ and that $m_0, M_0 \in \mathbb{R}$ with $m_0 \leq \inf_{x \in [a, b]} f(x) \leq \sup_{x \in [a, b]} f(x) \leq M_0$. Then $\exists c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = m_0 \int_a^c g(x)dx + M_0 \int_c^b g(x)dx.$$

3. Given a decreasing sequence of real numbers $\{G(n)\}$ such that $G(n) \rightarrow 0$ as $n \rightarrow \infty$. Define a function f on $[0, 1]$ in terms of $\{G(n)\}$ as follows:

$$f(x) = \begin{cases} 1, & \text{if } x = 0; \\ 0, & \text{if } x \text{ is irrational;} \\ G(n), & \text{if } x \text{ is the positive rational } \frac{m}{n} \text{ (in lowest terms).} \end{cases}$$

(a) (15 pt) Show that f is Riemann integrable on $[0, 1]$.

(b) (5 pt) Let $G(n) = \frac{1}{n}$. Define

$$g(x) = \begin{cases} 1, & \text{if } x \in (0, 1]; \\ 0, & \text{if } x = 0. \end{cases}$$

Show that $h(x) = g(f(x))$ is not Riemann integrable on $[0, 1]$. (Note that, in this case, f, g are both Riemann integrable on $[0, 1]$.)

4. (10 pt) Let f be continuous on a closed interval $[a, b]$ and let $M = \sup_{x \in [a, b]} |f(x)|$. Prove that

$$\lim_{p \rightarrow \infty} \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}} = M.$$